

Unravelling the multifarious role of wood vinegar made from waste biomass in plant growth promotion, biotic stress tolerance, and sustainable agriculture

Author links open overlay panelSumit Kumar ^{a b}, Mehjebin Rahman ^c, Ali Chenari Bouket ^d, Reza Ahadi ^e, [Mukesh Meena](#) ^{f 1}, Ingudam Bhupenchandra ^g, [Udai B. Singh](#) ^h, R. Arutselvan ⁱ, Ravindra Kumar ^j, Satyendra Pratap Singh ^a, Abhijeet S. Kashyap ^h, Ruchi Tripathi ^k, Sachin Gupta ^l, [Pranab Dutta](#) ^m, [Harish](#) ⁿ, Ramesh Singh ^o, Prashant Swapnil ^p

Show more

Share

Cite

<https://doi.org/10.1016/j.jaap.2024.106851>Get rights and content

Highlights

-
-

Pyroligneous acid (PA) is used as a biostimulant in agriculture

-
-

Wood vinegar contains beneficial compounds that significantly promote plant growth development.

-
-

PA promoted early seedling development and speed of germination in different crop plants.

-
-

PA triggered and encouraged the activation of transporter genes in root cells.

-
-

PA increases the metabolic activity of microorganisms in the decomposition or mineralization of P and S.

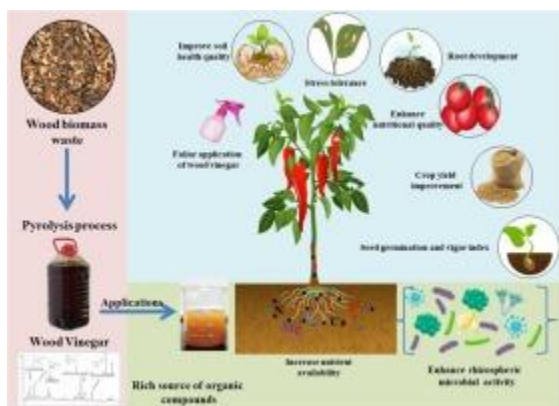
Abstract

The population of the world has increased intensively, and [sustainable agriculture](#) practices are important in achieving the zero-hunger goal.

The [agriculture](#) sector is constantly facing serious problems caused by [climate change](#) and the occurrence of pesticide-resistant causal agents. In this context, one of the most crucial

eco-friendly approaches, e.g., using plant-derived [bioproducts](#), is gaining more attraction because they have multifarious potential to overcome challenges. Wood [vinegar](#) (WV) or pyroligneous acid (PA) is a liquid biomaterial that is produced by the thermo-pyrolysis process of woody biomass and it was defined as a reddish-brown aqueous liquid that contains methanol, acetic acid, tars, and [wood oils](#) by the distillation process of woods. PA production and their application in agricultural fields, have attracted more attention recently because they may work as good bio-stimulants, [biopesticides](#), and [biofertilizers](#). Therefore, it may open a novel, promising window for [agriculture](#) and food production. PA has a significant role in elevating plant agronomic activities such as [seed germination](#) and vigor index, plant growth, nutritional value, and crop yields. This review aims to discuss in detail the following items: 1) The composition of wood [vinegar](#) and its production system; 2) Seed priming with the PA process; 3) Investigation of the role of PA in plant growth promotion activities, biocontrol potential, and its applications for alleviating [biotic stresses](#); 4) Describe the molecular mode of action of PA in suppressing plant diseases causal agents and promoting plant immunity through a well-illustrated diagram; 5) Evaluation of PA role in soil physicochemical, biological, and enzymatic activities and its impact on improving [compost quality](#) and curtailing emissions of green house gasses; 6) PA different advantages and limitations, as well as future perspectives for its usage and development.

Graphical Abstract



1. [Download: Download high-res image \(192KB\)](#)
2. [Download: Download full-size image](#)

Introduction

The global human population is increasing drastically to reach approximately 10 billion from the issues encountered nowadays, e.g., high birth rates till the year 2050, according to reports of FAO (Food and Agricultural Organization) [2], [3], [1]. The production of food

grains was 9.2 billion tonnes globally in 2018, which was around 50 % more in 2000, as a result of the rapid changes in demography, economic development, and lifestyle [4]. The population has increased by 34 % in several years, but the production of food crops is not increasing according to these circumstances [5]. Moreover, as the global population continues to increase, it will require more food production to fulfil its food demand. Only plants may directly provide human with 80 % of their protein and 90 % of their calorie requirements [6], [7]. Therefore, there is a need to produce 70–100 % more agricultural food grains from the current food grain production to feed the growing population and also equilibrate the socio-economic challenges that might arise due to population growth [8]. In the present state of affairs, the agriculture sector is facing several constraints, like biotic and abiotic stresses, that restrict its production and productivity. In addition, the day-by-day shortage of cultivated land and the impact of climate change are other threats to global food safety [9], [10].

Generally, plants face several biotic, soil, climatic, and anthropogenic threats, as illustrated in Fig. 1. These challenges are recognised as the main barrier to sustainable food grain production. Biotic threats mainly include microbial pathogens (Fungi, bacteria, viruses, and nematodes), weeds, and herbivores [11], [12], [13], [14]. Each year, biotic stresses cause 40 billion dollars losses in economy, which are responsible for 20–40 % of crop damages [15], [16]. Moreover, two major abiotic threats, e.g., salinity and drought, are belonging to soil and climatic threats, respectively. Salinity and drought are the most damaging and catastrophic challenges that affect plant growth and agricultural output. Salinity and drought are also responsible for causing billions of dollars in crop losses [17], [18], [19]. Heavy metal contamination is another severe worldwide problem for crop production, food security, and ecosystem [20], [21]. It has been reported that more than 70 % of heavy metals and their harmful by-products enter to the human body through contaminated food consumption [22]. For sustainable growth and development in agriculture, a huge dependency of producers on agrochemicals has been observed in recent days, which undoubtedly contributes to good results in agricultural output. According to Marlow et al. [23], 2.5 million metric tonnes of agrochemicals are applied annually. Due to superfluous and prolonged applications of synthetic chemicals, they have a long-term detrimental effect on human and animal health as well as cause soil nutrient depletion, water pollution, and environmental degradation [24], [26], [25]. If nothing is done to mitigate the notorious effects of excessive use of agrochemicals, they will destroy our natural ecosystem and cause noxious diseases in humans. Therefore, in the context of considering all these problems and according to the guidelines of the 2030 Agenda for sustainable development endorsed by the member states of the United Nations in 2015 [27], find out the innovative and eco-friendly technologies that can be safe for the

environment and can easily achieve sustainable crop production without causing any harmful effects. Hence, one of the most promising and emerging bio-based techniques is the use of natural biological materials such as wood vinegar (WV).

Wood vinegar (WV), or pyroligneous acid (PA), is a translucent reddish-brown aqueous liquid that is water-soluble and is produced through the process of condensation from material combustion [28], [29]. Meanwhile, it is a non-purified, concentrated, extraordinary, and oxygenated bio-liquid-based product that is produced by the thermo-pyrolysis process [30]. The term "pyrolysis" in older literature, primarily before the 1990s, referred to carbonization, focusing on biochar as the main product [31]. Modern pyrolysis, however, produces a range of products, including biochar, bio-oil, and syngas, depending on operating conditions [32]. The thermo-pyrolysis process is an eco-friendly process in which the thermal degradation of biomaterials at 400°C in the presence or absence of oxygen is used for the production of biomaterials such as biochar, wood vinegar, tar, etc. [33], [34]. Wood pyrolysis is a process that decomposes biomass under an inert environment in the absence of oxygen with a high temperature (200–600°C), resulting in main products, e.g., solid char (biochar, charcoal), tar, and by-product materials, e.g., non-condensable gases, and pyroligneous acid or wood vinegar [35]. Different types of pyrolysis include carbonization, methane pyrolysis, hydrous pyrolysis, dry distillation, destructive distillation, thermal depolymerization, ceramization, catagenesis, and flash vacuum pyrolysis [36]. Generally, materials that are used for the production of PA, such as rice husks, coconut shells, eucalyptus wood, oil palm tree branches, etc., contain various types of bio-compounds like cellulose, hemicellulose, and lignin [38], [39], [37]. PA is made up of 20 % organic material and the remaining 80 % water [40]. PA is composed of more than 200 biologically active compounds e.g., as syringol, catechol, acetol, propionic acid, methanol, butanol, phenolic acids, as well as mineral elements like Ca, Mg, Zn, Mn, and Fe [41], [42], [43], [44]. In PA, levoglucosan, cellobiosan, celotriosan, furans, cyclic ketones, glycolaldehyde, and acetic acid may be formed [45]. These compound compositions of PA mainly depend on some important factors such as the pyrolysis process used, heating rate, temperatures, moisture content in the feedstock, type of biomass used, particle size, and residence time [46], [47], [48]. Nowadays, due to their versatile nature, like being non-polluting and environmentally friendly, and as a kind of natural biomaterial, PA is increasingly used in different areas including the food industry, livestock, anti-inflammatory agents, household, agriculture, and medical areas [49], [50], [51].

In the agriculture sector, PA may work as an appreciated contributor to increasing sustainable agricultural crop production, food security, and protection of the environment. PA can efficiently play multifarious roles in promoting plant growth, enhancing the microbial diversity present in soil, and improving the chemical, physical, and biological

properties of soil [52]. PA is packed with various adorable features like biostimulators, antimicrobial agents, soil amendments, and foliar biofertilizers, and such features can ultimately induce crop yield and quality. Due to its acidic nature, PA is used to enhance nutrient availability and reduce nitric oxide (N₂O) and methane (CH₄) emissions [53]. Wood vinegar could help in the uptake and translocation of various macro- and micronutrients that assist plant growth and development [55], [54]. It was reported that PA has various crucial antifungal [56], antibacterial [57], insecticidal [58], herbicidal [59], and antitermitic [60] activities. PA applications protect field and vegetable crops from phytopathogen invasion by secreting various antimicrobial compounds [61], [62], [63], [64], [65]. It plays a main role in the mitigation of the adverse effects of environmental or abiotic stresses on crop plants and in preventing oxidative stress-related damage [66], [67], [68]. For instance, the application of PA in wheat may induce tolerance against drought stress by enhancing antioxidant activities, minimising ROS accumulation, and increasing abscisic acid production [69], [28].

Little is known about roles of PA in plant growth promotion, soil health restoration, and biotic stress alleviation. Moreover, availability and selection of the best-quality PA biomasses are major constraints for increasing the production and application of PA in sustainable agriculture. In this review, we will discuss in detail the PA production principles, their properties, and their potential role in plant growth promotion activities, highlighting the gaps that have yet to be filled. Moreover, we will further discuss the mechanisms of action of PA and elaborate on the crucial role of PA in the defence of plant against causal agents and mitigation strategies for biotic stresses and soil physicochemical, biological, and composting quality. Here we try to assist crop growers in reducing their reliance on agrochemicals while also providing profound perceptions and highlights on the environmental-friendly application of PA for achieving sustainable agriculture goals and building a greener agricultural system.

Wood vinegar and their properties

Pyroligneous acid is commonly referred to as wood acid. The words "pyrolysis" and "lignin," which make up a significant portion of plant biomass, are combined to form the term "pyroligneous acid" [47]. In older literature, the term "pyrolysis" typically refers to "carbonization", the primary output of which was biochar (solid char). Today, the term "pyrolysis" typically refers to operations where oils are the desired end product [70]. Bio-oil, liquid smoke, liquid oil, pyrolysis oil, liquid,

Various beneficial activities played by wood vinegar

Due to increasing population demands, farmers continuously use synthetic fertilizers on a large scale to achieve higher crop production without considering the harmful impact of synthetic fertilizers on human health, soil micro- and macroflora, and the ecosystem. There is a need to quickly adopt eco-friendly and bio-based techniques to easily enhance sustainable crop production without causing any undesirable side effects. In this aspect, one of the promising approaches is the application of

Wood vinegar and their role in biotic stress tolerance

Globally, the agriculture sector fulfills the food requirements of an exponentially increasing human population. To achieve successful food grain production, crop plants have a significant role. However, due to their sessile nature, plants cannot move and escape themselves from unwanted challenges. Plants encounter a range of biotic stressors throughout their lifecycle, spanning from germination to full maturity [275], [276]. According to an estimate, due to biotic stress factors, annually, 40

Application of wood vinegar as seed priming

The term "seed priming" was first given by Malnassy in 1971 [410], which refers to a technique that encourages the early and uniform emergence of seedlings [410]. Seed priming is a pre-sowing seed treatment technique that permits seeds to be hydrated under regulated conditions so they may take up water and go through the first stage of germination but prevents radical protrusion through the seed coat [412], [411]. After soaking, seeds are dried to their original moisture content. The seed goes

Wood vinegar and fermented bioextracts: natural products to enhance growth and yield

In the pursuit of sustainable agriculture, the use of natural products like wood vinegar and fermented bioextracts has gained significant attention due to their potential to enhance crop growth, improve yield, and mitigate the need for chemical inputs [144]. Both products have emerged as eco-friendly alternatives that can play a key role in promoting plant health while minimizing environmental impacts. Although wood vinegar and fermented bioextracts share common benefits, they also exhibit

Research gaps and conclusions

Pyrolysis of plant materials is an effective technique for the management of agricultural and agro-forestry waste residues. The result of pyrolysis is the formation of PA, in addition to other bio-based products such as biochar. PA is a potential mixture of various organic compounds with a wide range of applications in different fields. Recently, the utilization of PA has been studied intensively and identified in various aspects of agricultural sectors. Achieving sustainable development goals

Ethics approval

Not applicable.

Funding

This research was supported by the funding agency Science and Engineering Research Board (SERB), State University Research Excellence (SURE) (File Number: SUR/2022/005216) for this study.

Consent for publication

Not applicable.

Consent to participate

Not applicable.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The author M.M. gratefully acknowledges the financial support provided by the funding agency Science and Engineering Research Board (SERB), State University Research Excellence (SURE) (File Number: SUR/2022/005216) for this study. The author M.M. is also highly thankful to the Ministry of Education and SPD-RUSA Rajasthan for the financial support received under the RUSA-2.0 project. All the authors acknowledge their host institute for infrastructure support. All the authors read and approved